FINAL YEAR PROJECT (FYP)
PROPOSAL DOCUMENT



PROJECT TITLE

TourGB (AI-Based Trip Planning, Accommodation Booking, and Intra-GB Transport Reservation System for Gilgit-Baltistan)

BS (IT)

SESSION: 2023-2027

GROUP MEMBERS

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1 Introduction

TourGB is a web-based tourism platform for Gilgit-Baltistan that helps tourists plan trips, find accommodation, and reserve Intra-GB transport in one place. Tourists can browse destinations and attractions, use manual search and filters, or use a conversational AI assistant to describe their travel requirements and receive a trip plan.

The system also allows local accommodation and transport providers to register their services, manage prices and availability, and receive reservations after administrator approval. Tourists can manage their bookings and receive system notifications.

1.1Project Title

"TourGB (AI-Based Trip Planning, Accommodation Booking, and Intra-GB Transport Reservation System for Gilgit-Baltistan)"

1.2 Project Overview Statement

Planning a trip to Gilgit-Baltistan can require tourists to use different sources for destinations, accommodation, local transport, and travel information. Searching separately for suitable services, prices, availability, and activities can take time and make planning difficult.

TourGB brings trip planning, accommodation search and booking, and Intra-GB transport search and reservation into one platform. Tourists can search manually using filters or use the conversational AI assistant to provide their requirements and receive suitable recommendations and a day-by-day travel plan.

The platform also provides destination and attraction information, weather information, and available road-status information to support travel decisions and allows tourists to manage their reservations.

Accommodation and transport providers can register services, manage prices and availability, and receive reservations after administrator approval.

1.3 Project Overview Statement Template

Project Title: TourGB (AI-Based Trip Planning, Accommodation Booking, and Intra-GB Transport Reservation System for Gilgit-Baltistan)

Group Leader: Nadir Ahmad

Project Members:

Name	Roll no. #	Email Address	Signature
Nadir Ahmad	102613	nadirahmad5423@gmail.com	
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Project Goal:

To provide tourists with one web-based platform for trip planning, accommodation booking, and Intra-GB transport reservation, while also providing useful destination, weather, and road-status information.

Objectives:

Sr. No.	
1	Develop a conversational AI assistant that collects trip requirements and generates a day-by-day travel plan using approved TourGB information.
2	Provide public browsing and manual search, filtering, booking, and reservation for accommodation and Intra-GB transport.
3	Allow accommodation and transport providers to register and manage their services, prices, availability, and reservations after administrator approval.
4	Provide weather and available road-status information and warn tourists when a reported blockage affects a planned or booked route. Map-based road information will be the primary source, while administrators can add or update advisories when map information is missing or outdated.

Project Success criteria:

When our system is complete and running, tourists can browse destinations and services, plan trips, and search and reserve accommodation and Intra-GB transport. The AI assistant can generate a day-by-day plan from the tourist's requirements, and users can manage their reservations and receive travel warnings when available road or weather information affects their trip.

Assumptions, Risks, and Obstacles:

1- Providers are expected to keep service details, prices, and availability updated.

2- Map-based road information is the primary source for available road-status information; administrators can add or update an advisory when information is missing or outdated.

3- Weather information depends on the selected weather data source.

Organization Address (if any):

Department of Information Technology, Akhuwat College Kasur, affiliated with University of the Punjab, Lahore.

Type of project: Research ●Development

Target End users:

1- Tourists planning or visiting Gilgit-Baltistan. (Primary users)

2- Service providers and administrator. (Secondary/system users)

Development Technology: ● Object Oriented Structured

Platform: ● Web Based

Suggested Project Supervisor: Mr. Shafiq-u-Rehman

Approved By:**Mr. Shafiq-u-Rehman**

Date: September 29, 2026

1.4 Project Goals & Objectives

Goals

Our main goal is to develop a web-based platform for Gilgit-Baltistan where tourists can plan trips, find and reserve accommodation and Intra-GB transport, and receive useful tourism, weather, and road-status information in one place.

Objectives

- Develop a conversational AI assistant that collects trip requirements and generates a day-by-day trip plan using TourGB data.
- Provide manual search, filtering, and reservation for accommodation and Intra-GB transport.
- Allow accommodation and transport providers to register, manage their services, prices, availability, and reservations after administrator approval.
- Provide weather information and available road-status information, using map-based road information as the primary source and administrator updates as a secondary source when needed.

1.5 High-level system components

Generally, a web application is composed of many different components that work collectively to address the problem for which the system has been devised. The high-level system components / modules for our application are the following:

- Authentication and User Management
- Tourist Discovery and Search
- Booking and Reservation Management
- Service Provider Management
- AI Assistant and RAG System
- Road Advisory and Weather Safety
- Email Notification System
- Administrator Panel
- Online Payment Integration

1.6 List of optional functional units

- Urdu Language Interface
- Mobile Application

1.7 Exclusions

Flight booking, real-time GPS vehicle tracking, government permit processing, direct Booking.com/Airbnb integration, and production-scale commercial deployment are outside the FYP scope.

1.8 Application Architecture

We will follow 3-tier Architecture.

- Next.js (Front End)
- Django + Django REST Framework + LangChain + Local AI Model (Back End)
- PostgreSQL + PostGIS + pgvector (Database)

1.9 Gantt chart

Sr. No.	**Task Name**	**Start Date**	**End Date**	**Duration (Days)**
1.	Project Proposal	01-10-2026	15-10-2026	15
2.	Requirements Analysis and Literature Review	16-10-2026	15-11-2026	31
3.	Software Requirements Specification (SRS)	16-11-2026	15-12-2026	30
4.	System and Database Design (SDS)	16-12-2026	31-01-2027	47
5.	GUI Design and Working Prototype	01-02-2027	28-02-2027	28
6.	Software Development	01-03-2027	30-04-2027	61
7.	AI/RAG and Integration	01-04-2027	15-05-2027	45
8.	Testing and Bug Fixing	16-05-2027	15-06-2027	31
9.	GUI Refinements and Final Integration	16-06-2027	30-06-2027	15
10.	Deployment and Final Documentation	01-07-2027	31-07-2027	31

1.9 Gantt Chart

The following Gantt chart shows the planned schedule for the development of TourGB. The project will be completed in 10 months, from October 2026 to July 2027. The first 5 months will be used for the proposal, requirements, design and prototype. The next 5 months will cover the development of all modules, testing, deployment and documentation.

Sr. No.	Task / Module	Start Date	End Date	Duration (Days)
1	Project Proposal	01-10-2026	15-10-2026	15
2	Requirements Analysis and Literature Review	16-10-2026	15-11-2026	31
3	Software Requirements Specification (SRS)	16-11-2026	15-12-2026	30
4	System and Database Design (SDS)	16-12-2026	31-01-2027	47
5	GUI Design and Working Prototype	01-02-2027	28-02-2027	28
6	M1 - Authentication and User Management	01-03-2027	20-03-2027	20
7	M8 - Administrator Panel	10-03-2027	31-03-2027	22
8	M4 - Service Provider Management	21-03-2027	20-04-2027	31
9	M6 - Road Advisory and Weather Safety	21-03-2027	10-04-2027	21
10	M2 - Tourist Discovery and Search	01-04-2027	30-04-2027	30
11	M3 - Booking and Reservation Management	10-04-2027	20-05-2027	41
12	M7 - Email Notification System	01-05-2027	15-05-2027	15
13	M5 - AI Assistant and RAG System	21-04-2027	15-06-2027	56
14	Full System Integration and Testing	16-06-2027	30-06-2027	15
15	Deployment and Final Documentation	01-07-2027	31-07-2027	31



Figure 1.1: Gantt Chart for TourGB Development (October 2026 – July 2027)

1.10 Hardware and Software Specification

Hardware: (Recommended)

Processor	Intel Core i5 (8th generation or newer)
RAM	16 GB recommended for acceptable local AI model performance. 8 GB is the minimum.
Storage	256 GB SSD minimum. Local AI model file is approximately 4-5 GB.
Internet	Stable broadband for development and weather API access.

Software: (Recommended)

OS	Windows 10/11 (64-bit) or Linux Ubuntu 22.04+
Python 3.11+	For Dajngo, LangChain, PostgrSql, and Ollama
PostgreSQL 15+	Primary relational database
VS Code + Git	Code editor and version control

1.11 Tools and Technologies Used with Reasoning

Tools:

Visual Studio Code

Visual Studio Code (VS Code) is an open-source and cross-platform code editor. It will be used for developing the frontend, backend, database-related code, and AI integration of TourGB.

Git/GitHub

Git/GitHub are used for version control and project collaboration. They will be used to store the project source code, track changes, and allow both team members to work on the same project.

Postman

Postman is a tool used for testing APIs. It will be used to send requests to the Django REST APIs and check their responses during backend development.

Ollama

Ollama is a tool for running language and embedding models locally. It will be used to run lightweight AI models on the team's computers without depending on a paid cloud AI API.

Technology:

Next.js

Next.js is a React-based framework used to build the TourGB web frontend. It will be used for public tourism pages, accommodation and transport pages, the AI chatbot, tourist dashboard, provider dashboard, and admin interface.

TypeScript

TypeScript is a programming language based on JavaScript that adds type checking to the code. It will be used with Next.js to make the frontend code more organized and easier to maintain.

Django

Django is a Python web framework used for backend development. It will handle authentication, provider approval, booking rules, data validation, business logic, and communication with the database.

Django REST Framework

Django REST Framework is a toolkit for building web APIs with Django. It will be used to provide APIs that connect the Next.js frontend with the Django backend.

Python

Python is a high-level programming language used for backend and AI development. It will be used with Django, LangChain, and the local AI components of TourGB.

PostgreSQL

PostgreSQL is an open-source relational database management system. It will be used to store users, service providers, accommodations, transport services, bookings, attractions, reviews, road information, and other TourGB data.

PostGIS

PostGIS is an extension of PostgreSQL that adds geographic data support. It will be used to store coordinates and perform location-based operations such as distance calculation and finding nearby accommodations, attractions, and transport services.

pgvector

pgvector is a PostgreSQL extension for storing and searching vector embeddings. It will be used to store embeddings of tourism documents and perform similarity search for the RAG system.

LangChain

LangChain is a framework used to build AI applications. It will be used to organize the RAG pipeline and connect the local AI model with the knowledge base, database information, weather data, and other TourGB functions.

Local AI Language and Embedding Models(Ollama and Qwin)

Lightweight local language and embedding models will be used for the conversational AI and RAG system.

Open-Meteo API

Open-Meteo is a weather API that provides weather forecasts using location coordinates. It will be used to display weather information for selected destinations and travel dates.

Leaflet.js and OpenStreetMap

Leaflet.js is a JavaScript library used to display interactive maps, while OpenStreetMap provides map data. They will be used to display destinations, attractions, accommodations, transport locations, routes, and available road information on the TourGB map.

Django Email / SMTP

Django's email system with SMTP will be used to send account and booking-related emails. These include registration, provider approval, booking confirmation, cancellation, and road-related notifications.